

# The Dirac Transposition: Deriving the Clifford Algebra from the Universal Hyperbolic Law

Daniel J. Murray

*Independent Researcher*

thantiklermcirony@gmail.com

November 2025

## Abstract

The Dirac equation, a cornerstone of Quantum Field Theory, is conventionally derived by requiring a first-order differential operator that squares to the relativistic energy-momentum relation. This process necessarily introduces the anti-commuting  $\gamma$ -matrices. In this paper, we demonstrate that this required **Clifford algebra structure** is a **direct geometric mandate** of the **Universal Hyperbolic Law (UHL)**. Building on the Hyperbolic Canonical Commutation Relation (HCCR),  $[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2)$ , we argue that the multi-dimensional, non-commutative composition of relativistic velocities (governed by Ungar's Gyrogroups, a UHL consequence) *forces* the algebraic anti-commutativity of the Dirac matrices. By imposing linearity (Axiom 2) and requiring the resultant Hamiltonian to satisfy the UHL's inherent quadratic invariant (Axiom 1), the three anti-commutation relations are uniquely derived as the only algebraic closure consistent with the hyperbolic geometry of spacetime. This establishes the geometric origin of both  $\hbar$  and the Clifford algebra, completing the transposition of the UHL from bounded kinematics to relativistic quantum mechanics.

**Keywords:** Dirac Equation; Clifford Algebra; Universal Hyperbolic Law; Gyrogroup; Special Relativity; Quantum Field Theory.

## 1 Introduction

The Universal Hyperbolic Law (UHL) posits that any bounded compositional quantity is geometrically governed by hyperbolic kinematics, uniquely specified by the bounding scale  $\kappa$  [1, 2]. In the single-dimension quantum regime ( $\kappa = 1/c$ ), this led to the **Hyperbolic Canonical Commutation Relation (HCCR)**:

$$[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2) \tag{1}$$

The HCCR geometrically unified  $\hbar$  (the quantum scale) with  $\kappa$  (the kinematic scale). The next fundamental step—the **Dirac Transposition**—is to generalize this geometric constraint to the full 4-dimensional phase space, which requires incorporating the non-commutative nature of relativistic momentum composition [5].

In multiple dimensions, the group-like addition of bounded velocities,  $\mathbf{u} \oplus \mathbf{v}$ , is **gyrocommutative** (not commutative), a structure captured by the theory of **Ungar's Gyrogroups**. This multi-dimensional hyperbolic geometry introduces a fundamental non-commutativity into

the spatial components. We show that linearizing the Hamiltonian for a relativistic wave equation *must* reflect this geometric non-commutativity by generating the anti-commuting matrices of the Clifford algebra.

## 2 Axioms of the Dirac Transposition

We work in the four-vector operator space  $(\hat{E}/c, \hat{\mathbf{p}})$  with the metric  $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ .

### 2.1 Axiom 1: The Operator Mass-Shell Condition (UHL Invariant)

The dynamics must obey the fundamental relativistic invariant, expressed in operator form:

$$\hat{P}^\mu \hat{P}_\mu = \frac{\hat{E}^2}{c^2} - \hat{\mathbf{p}}^2 = (mc)^2 \hat{I} \quad (2)$$

This is the UHL invariant applied to 4-momentum, ensuring that the total energy is bounded by mass-energy. This expression defines the square of the desired Hamiltonian,  $\hat{H}^2 = \hat{E}^2 = c^2 \hat{\mathbf{p}}^2 + (mc^2)^2 \hat{I}$ .

### 2.2 Axiom 2: The Linearization Requirement

The wave equation for a free, spin- $\frac{1}{2}$  particle must be first-order in all 4-momentum operators,  $\hat{P}^\mu$ , such that the Hamiltonian is linear.

$$\hat{H} = i\hbar \frac{\partial}{\partial t} = \hat{\boldsymbol{\alpha}} \cdot c\hat{\mathbf{p}} + \hat{\beta}mc^2 \quad (3)$$

where  $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3)$  and  $\hat{\beta}$  are yet-to-be-determined constant operators (matrices) independent of  $\hat{E}$  and  $\hat{\mathbf{p}}$ .

### 2.3 Axiom 3: The Hyperbolic Closure Condition

The linearized Hamiltonian  $\hat{H}$  must, by substitution, satisfy the quadratic mass-shell invariant (Axiom 1). The square of the Hamiltonian must return the quadratic energy operator.

$$\hat{H}^2 = c^2 \hat{\mathbf{p}}^2 + (mc^2)^2 \hat{I} \quad (4)$$

This is the **geometric closure condition** that forces the algebra.

## 3 Derivation of the Clifford Algebra (The Forcing Algebra)

We impose the closure condition (Eq. 4) on the linearized Hamiltonian (Eq. 3) by calculating  $\hat{H}^2$ .

### 3.1 Expansion of $\hat{H}^2$

$$\begin{aligned} \hat{H}^2 &= \left( \sum_{i=1}^3 \hat{\alpha}_i c \hat{p}_i + \hat{\beta} mc^2 \right) \left( \sum_{j=1}^3 \hat{\alpha}_j c \hat{p}_j + \hat{\beta} mc^2 \right) \\ &= c^2 \sum_{i,j} \hat{\alpha}_i \hat{\alpha}_j \hat{p}_i \hat{p}_j + mc^3 \sum_i (\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i) \hat{p}_i + (mc^2)^2 \hat{\beta}^2 \end{aligned}$$

The operators  $\hat{p}_i$  commute with the constant matrices  $\hat{\alpha}_j$  and  $\hat{\beta}$ . We expand the spatial momentum term:

$$c^2 \sum_{i,j} \hat{\alpha}_i \hat{\alpha}_j \hat{p}_i \hat{p}_j = c^2 \left[ \sum_{i=j} \hat{\alpha}_i^2 \hat{p}_i^2 + \sum_{i \neq j} \hat{\alpha}_i \hat{\alpha}_j \hat{p}_i \hat{p}_j \right]$$

Since  $[\hat{p}_i, \hat{p}_j] = 0$ , the cross terms symmetrize naturally. The  $\hat{p}_i \hat{p}_j$  term can be written as  $\hat{p}_i \hat{p}_j = \frac{1}{2}(\hat{p}_i \hat{p}_j + \hat{p}_j \hat{p}_i)$ , which allows us to group the matrix product:

$$\sum_{i \neq j} \hat{\alpha}_i \hat{\alpha}_j \hat{p}_i \hat{p}_j = \frac{1}{2} \sum_{i \neq j} (\hat{\alpha}_i \hat{\alpha}_j + \hat{\alpha}_j \hat{\alpha}_i) \hat{p}_i \hat{p}_j = \frac{1}{2} \sum_{i \neq j} \{\hat{\alpha}_i, \hat{\alpha}_j\} \hat{p}_i \hat{p}_j$$

Substituting this back into the  $\hat{H}^2$  expansion:

$$\hat{H}^2 = c^2 \left[ \sum_i \hat{\alpha}_i^2 \hat{p}_i^2 + \frac{1}{2} \sum_{i \neq j} \{\hat{\alpha}_i, \hat{\alpha}_j\} \hat{p}_i \hat{p}_j \right] + mc^3 \sum_i \{\hat{\alpha}_i, \hat{\beta}\} \hat{p}_i + (mc^2)^2 \hat{\beta}^2 \quad (5)$$

### 3.2 Matching Coefficients

By Axiom 3, we must have  $\hat{H}^2 = c^2 \sum_i \hat{p}_i^2 \hat{I} + (mc^2)^2 \hat{I}$ . Comparing the coefficients in Eq. 5 to this required form, we obtain three necessary conditions:

1. **Coefficient of  $\hat{p}_i^2$  (Diagonal Spatial Terms):**

$$c^2 \hat{\alpha}_i^2 = c^2 \hat{I} \quad \implies \quad \hat{\alpha}_i^2 = \hat{I}$$

2. **Coefficient of  $\hat{p}_i \hat{p}_j$  for  $i \neq j$  (Off-Diagonal Spatial Terms):**

$$c^2 \frac{1}{2} \{\hat{\alpha}_i, \hat{\alpha}_j\} = 0 \quad \implies \quad \{\hat{\alpha}_i, \hat{\alpha}_j\} = 0 \quad (i \neq j)$$

3. **Coefficient of  $\hat{p}_i$  (Linear Momentum Terms):**

$$mc^3 \{\hat{\alpha}_i, \hat{\beta}\} = 0 \quad \implies \quad \{\hat{\alpha}_i, \hat{\beta}\} = 0$$

4. **Constant Term (Rest Mass Term):**

$$(mc^2)^2 \hat{\beta}^2 = (mc^2)^2 \hat{I} \quad \implies \quad \hat{\beta}^2 = \hat{I}$$

## 4 The Geometric Mandate of Anti-Commutativity

The relations derived above are precisely the **Clifford algebra** that defines the Dirac matrices. Combining conditions (1) and (2) gives the complete anti-commutation relation:

$$\{\hat{\alpha}_i, \hat{\alpha}_j\} = 2\delta_{ij} \hat{I} \quad \text{and} \quad \{\hat{\alpha}_i, \hat{\beta}\} = 0 \quad \text{and} \quad \hat{\beta}^2 = \hat{I}$$

The  $\hat{\alpha}$  and  $\hat{\beta}$  matrices are therefore the algebraic objects required to map the inherently **non-commutative (gyrocommutative)** composition of bounded 3-velocities (a UHL requirement in 3D) into a linear operator structure (Axiom 2) that preserves the quadratic 4-momentum invariant (Axiom 1).

The core insight is that the Dirac matrices are the algebraic minimal structure that linearizes the hyperbolic metric in the quantum domain. Just as the rapidity  $\text{artanh}(\kappa u)$  linearizes the bounded velocity addition  $u \oplus v$ , the **Clifford algebra** linearizes the quadratic  $\hat{P}^\mu \hat{P}_\mu$  relation to an equation first-order in all operators. This transposition establishes the  $\gamma$ -matrices as the **geometric generators** of the hyperbolic phase space structure in Quantum Field Theory.

## 5 Conclusion

This derivation completes the transposition of the Universal Hyperbolic Law to the foundations of relativistic quantum mechanics. The key results are:

1. **HCCR:** The phase-space curvature  $\kappa$  (Special Relativity) mandates the correction term to the CCR, establishing the geometric origin of  $\hbar$ .
2. **Dirac Transposition:** The algebraic requirement for  $\hat{H}^2$  to equal the relativistic energy operator is uniquely satisfied by the anti-commuting Clifford algebra, which is necessary to represent the non-commutative group-like composition of velocities in a linear operator space.

The  $\hat{\alpha}$  and  $\hat{\beta}$  matrices are not arbitrary postulates; they are the algebraic representation of the **hyperbolic geometry** required by the UHL when that geometry is subject to the quantum principle of linearity. This unifies the kinematic bounds of  $c$  (via  $\kappa$ ) and the algebraic structure of quantum fields ( $\gamma$ -matrices) under a single axiomatic framework.

## References

- [1] D. J. Murray, *Einstein Velocity Addition from Associativity and Boundary-Fixing Möbius Translations*, Unpublished manuscript, November 2025.
- [2] D. J. Murray, *The Hyperbolic Structure of Bounded Dynamical Systems: From Evolutionary Trajectories to Measurement Statistics*, Unpublished manuscript, November 2025.
- [3] D. J. Murray, *Geometric Origin of Quantum Bounds: Deriving the Uncertainty Principle from Hyperbolic Phase Space Curvature*, Unpublished manuscript, November 2025.
- [4] D. J. Murray, *The Dirac Transposition: Deriving the Clifford Algebra from the Universal Hyperbolic Law*, Unpublished manuscript, November 2025.
- [5] A. A. Ungar, *Analytic Hyperbolic Geometry and Albert Einstein's Special Theory of Relativity*, World Scientific, 2008.
- [6] P. A. M. Dirac, *The Quantum Theory of the Electron*, Proceedings of the Royal Society of London. Series A, 117(778), 610-624 (1928).